

Unit 3.2 Probability

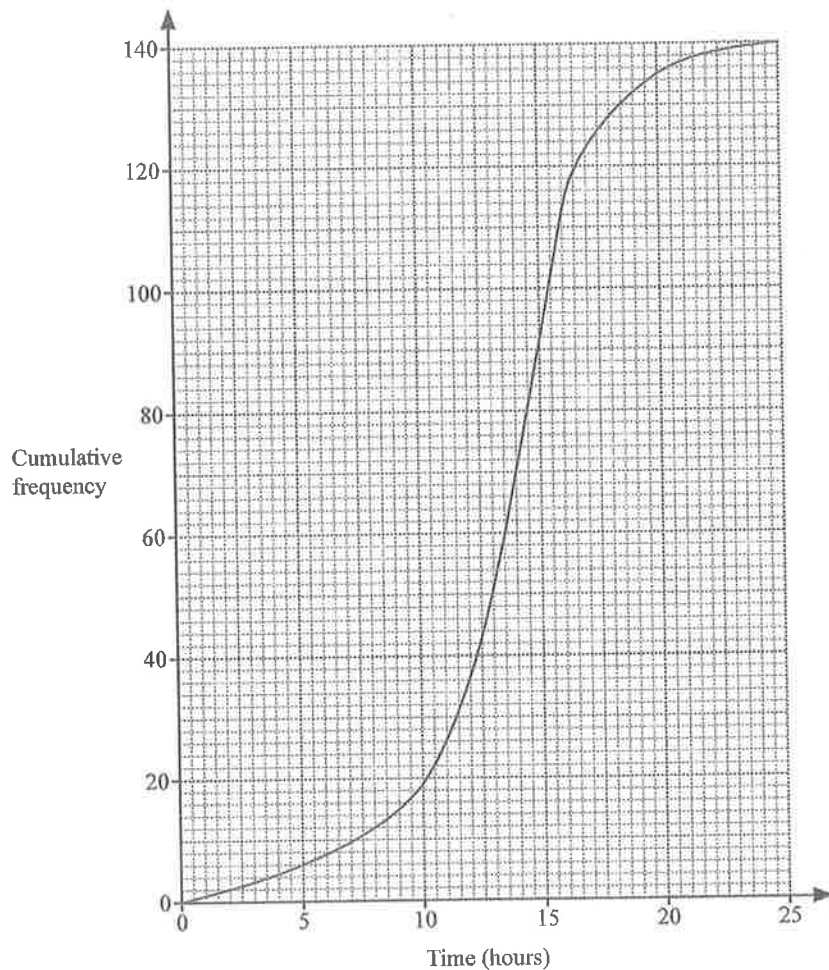
Paper 1: Basic

1. (a) A bag contains pink counters, blue counters and yellow counters.
A counter is picked at random from the bag.
The probability that the counter is blue is $\frac{2}{5}$.
The probability that the counter is yellow is $\frac{2}{15}$.
Find the probability that the counter is pink. [1]
- (b) Another bag contains 8 red counters and 16 green counters.
More red counters are added to the bag.
The probability of picking a green counter from this bag at random is now $\frac{1}{4}$.
Find the number of red counters added to the bag. [2]
[N22/I/5]
2. A bag contains some red marbles, some white marbles and some yellow marbles.
The probability of picking a red marble at random is 0.44.
The probability of picking a white marble at random is 0.24.
(a) Find the probability of picking a yellow marble at random. [1]
(b) There are 175 marbles in the bag.
Find the number of white marbles in the bag. [1]
[N21/I/4]
3. A bag contains some red counters, some blue counters and some yellow counters.
The probability of choosing a red counter at random is 0.45.
The probability of choosing a blue counter at random is 0.3.
(a) Find the probability of choosing a yellow counter at random. [1]
(b) In the bag there are 9 more red counters than blue counters.
Find the total number of counters in the bag. [2]
[N20/I/8]
4. A bag contains 8 red counters, 9 green counters and 3 yellow counters.
(a) A counter is chosen at random and then replaced.
What is the probability that it is **not** a yellow counter? [1]
(b) x green counters are removed from the bag.
The probability of choosing a red counter is now $\frac{2}{3}$.
Find the value of x . [1]
[N19/I/7]

5. The frequency table shows information about the time, in hours, that each of 140 adults watched television in one week.

Time (t hours)	$0 < t \leq 5$	$5 < t \leq 10$	$10 < t \leq 13$	$13 < t \leq 16$	$16 < t \leq 20$	$20 < t \leq 25$
Frequency	6	13	32	62	22	5

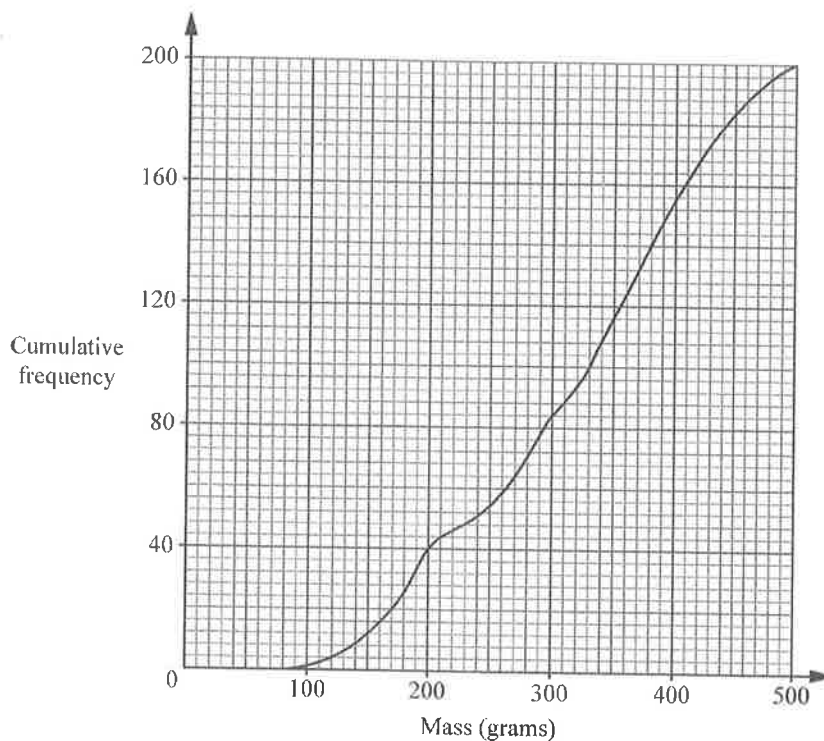
- (a) Calculate an estimate of the mean time. [1]
 (b) This cumulative frequency diagram shows the same information.



- (i) 40% of the adults watch more than n hours each week.
 Find n . [2]
 (ii) One of the adults is chosen at random.
 Find the probability that the adult watches between 8 and 13.5 hours. [2]

[N20/1/23]

6. Some students were asked to estimate the mass, in grams, of sweets in a jar.
The cumulative frequency diagram shows the results.



The actual mass of the sweets is 300 grams.

- (a) Find the probability that a student, chosen at random, overestimated the mass. [2]
(b) Find the number of students who gave estimates within 10% of the actual mass. [2]

[N16/I/21]

7. A bag contains 10 marbles, n of which are red and the rest are yellow.
A marble is chosen at random and not replaced.

- (a) Write down, in terms of n , the probability that the marble is yellow. [1]

A second marble is chosen at random.

- (b) Find, in terms of n , the probability that both marbles are yellow. [1]

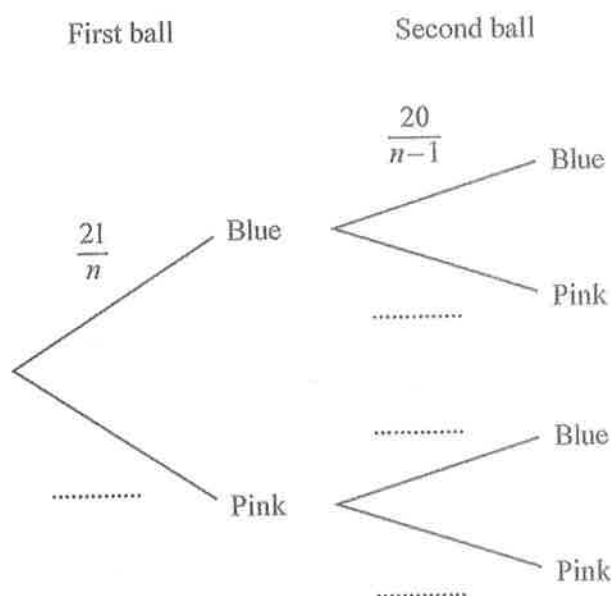
- (c) (i) The probability that both marbles are yellow is $\frac{1}{15}$.

Show that $n^2 - 19n + 84 = 0$. [2]

- (ii) Solve the equation $n^2 - 19n + 84 = 0$ to find the number of yellow marbles in the bag. [3]

[N15/I/24]

1. A bag contains n balls.
 21 of the balls are blue and the rest are pink.
 Leon takes two balls from the bag, at random, without replacement.
 (a) Complete the tree diagram.



[2]

- (b) The probability that Leon takes two pink balls is $\frac{1}{8}$.
 Write down an equation to represent this information and show that it simplifies to

$$n^2 - 49n + 528 = 0.$$

 (c) Solve the equation $n^2 - 49n + 528 = 0$.
 (d) Explain why one of the solutions in part (c) must be rejected.
 (e) Find, as a fraction in its simplest form, the probability that Leon takes one blue ball and one pink ball.

[3]

[3]

[1]

[2]

[N20/II/9]

2. Jiao is researching maternity care in a hospital.
 She searches the internet to find some information.
 This table gives information about the ages of the mothers of the babies born in the hospital on one day and the sexes of their babies.
 Each of the mothers gave birth to only one baby.

	Mother aged under 30	Mother aged 30 or over
Boy	3	9
Girl	7	6

- (i) One of these mothers is selected at random.
 Find, as a fraction in its lowest terms, the probability that she is under 30.
 (ii) Two of the mothers are selected at random.
 Find the probability that
 (a) they both have a baby girl,
 (b) they are both aged 30 or over, but only one has a baby boy.

[1]

[2]

[2]

[N18/II/9(b)]

3. The table summarises the number of passengers in each of the cars in the morning.

Number of passengers	0	1	2	3	4
Number of cars	34	17	13	11	5

- (i) One of the cars is selected at random.

Find the probability that it has no passengers.

[1]

- (ii) Two of the cars are selected at random.

Find, as a fraction in its simplest form, the probability that

- (a) they both had four passengers,

[2]

- (b) one had more than two passengers and the other had fewer than two passengers.

[2]

[N17/II/9(b)]

4. A bag contains 9 white balls, 5 red balls and 2 black balls.

Two balls are taken from the bag without replacement.

- (i) Draw a tree diagram to show the probabilities of the possible outcomes.

[2]

- (ii) Find, as a fraction in its simplest form, the probability that

- (a) the two balls are the same colour,

[2]

- (b) at least one of the balls is white.

[2]

[N16/II/9(b)]

Paper 2: Intermediate

5. The table shows the languages studied by a group of 30 students.

	French	Not French
Spanish	8	12
Not Spanish	7	3

- (i) One of the students who studies French is chosen at random.

Find the probability that this student also studies Spanish.

[1]

- (ii) Two of the students who study Spanish are chosen at random.

Find the probability that both students study Spanish but not French.

[2]

- (iii) Three students are chosen at random from the whole group.

Find the probability that only one of them studies Spanish.

[2]

[N22/II/9(b)]

6. A small business has 16 employees.

One of the 16 employees is selected at random.

The probability that it is a man working part time is $\frac{1}{8}$.

Two of the 16 employees are selected at random.

The probability that they are both women working full time is $\frac{1}{8}$.

Complete the table of information about the 16 employees of the business.

	Part-time workers	Full-time workers
Men		5
Women		

[4]

[N21/II/9(b)]

7. A bag contains 16 red counters and 9 blue counters.
Two counters are taken from the bag at random without replacement.

(i) Shen says the probability that both counters are blue is $\frac{81}{625}$.

Explain what he has done wrong.

[1]

(ii) Draw a tree diagram to show the probabilities of the possible outcomes.

[2]

(iii) Find, as a fraction in its simplest form, the probability that only one of the counters is red.

[2]

[N19/II/9(b)]

8. The table shows the ages of the 240 people who entered the cycle race.

Age (a years)		$20 \leq a < 30$	$30 \leq a < 40$	$40 \leq a < 50$	$50 \leq a < 60$
Frequency	Men	35	46	24	15
	Women	27	34	41	18

- (i) One of these people is selected at random.

Find, as a fraction in its lowest terms, the probability that the person is

(a) a man aged 50 or more,

[1]

(b) aged under 30.

[1]

- (ii) Two of these people are selected at random.

Find the probability that **both** of them are women aged under 40.

Give your answer as a decimal correct to three significant figures.

[2]

[N15/II/10(b)]